

Concurrent Programming - Exercícios 2

CCS sequential: CCS_0^ω

1. Solve the exercises of CCS_0 in <https://pseuco.com/#/exercises> (Pseuco.com)
2. Let $Act = \{a, b, c\}$, compute using the inference system:
 - (a) $\llbracket a.b.0 + 0 \rrbracket$
 - (b) $\llbracket a.(b.0 + 0) \rrbracket$
 - (c) $\llbracket a.b.c.0 + b.(0 + a.0) \rrbracket$

3. Consider the following definition of a coffee machine

$$CM := \text{coin.coffee}.CM$$

- (a) Compute Γ and $\llbracket CM \rrbracket_\Gamma$ and implement it in pseuco.com
 - (b) Write a process that behaves as CM can steal the coin, i.e., does not give coffee.
 - (c) Write a process that behaves as CM but can give coffee or tea. Consider two cases: the beverage is randomly chosen or there is a way to ask the buyer if he wants coffee or tea.
 - (d) Write a process that behaves as CM but can give tea for 0.5 euros and coffee for 1 euro.
 - (e) Repeat (a) for the last questions.
4. Let

$$\begin{aligned} \Gamma = \{ & (P, a.P_1), (P_1, b.P + c.P), (Q, a.Q_1) \\ & (Q_1, b.Q_2 + c.Q), (Q_2, a.Q_3), (Q_3, b.Q + c.Q_2) \} \end{aligned}$$

Write the equations defined by Γ . Using the inference system $\longrightarrow_\Gamma$, compute $\llbracket P \rrbracket_\Gamma$ e $\llbracket Q \rrbracket_\Gamma$. Draw the diagrams. Implement in pseuco.com.

5. For each expression and set of equations, indicate the set Γ and the semantics of each expression using $\longrightarrow_\Gamma$.
 - (a) $\llbracket A \rrbracket_\Gamma$ being $A := a.(b.0 + b.c.A)$
 - (b) $\llbracket B \rrbracket_\Gamma$ being $A := a.A + \tau.b.A$ and $B = a.A + b.A$
 - (c) $\llbracket C \rrbracket_\Gamma$ being

$$\begin{aligned} C &:= c.C + D \\ D &:= 0 + c.C \end{aligned}$$

- (d) $\llbracket C_0 \rrbracket_\Gamma$ being

$$\begin{aligned} C_0 &:= \text{inc}.C_1 \\ C_n &:= \text{inc}.C_{n+1} + \text{dec}.C_{n-1}, \text{ para } n \geq 1 \end{aligned}$$

- (e) $\llbracket X \rrbracket_\Gamma$ and $\Gamma = \{(X, X + 0)\}$

6. Say if A and B are guarded or unguarded: $A := a.A + B$ and $B := b.B + A$.
7. Say which variables are guarded in the following equations: $C := c.C + D$, $D := 0 + c.C$
 $A := b.0 + A$ e $B := b.B + a.A$.
8. Solve the exercices in PseuCo Book (<https://book.pseuco.com/>) for sequential CCS without and with recursion.

Inference Rules CCS_0^ω

$$\text{prefix} \frac{}{\alpha.P \xrightarrow{\alpha} P}$$

$$\text{choiceL} \frac{P \xrightarrow{\alpha} P'}{P + Q \xrightarrow{\alpha} P'}$$

$$\text{choiceR} \frac{Q \xrightarrow{\alpha} Q'}{P + Q \xrightarrow{\alpha} Q'}$$

$$\text{Rec} \frac{P \xrightarrow{\alpha} P' \quad \Gamma(X) = P}{X \xrightarrow{\alpha} P'}$$